UNIT 2 · STANDARD 6.DS.3

Describe Data by Range and Interquartile Range

Lesson 2-5

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the range and the interquartile range of a data set and use them to describe how spread out the data is.

Language Objective: I can explain what the spread of a data set is using the words range, quartile, interquartile range, and vary.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Range and Interquartile Range | **Range** | **Quartile** | **Interquartile range** | **Variability** | **Measure of variation** | **Extreme value**

Level 3: try the blanks from memory first, then check the bank.

1 The range uses the _____ values; the interquartile range uses the _____.

2 Subtract the least value from the _____ value to find the range.

3. Numbers that split ordered data into four equal parts are _____.

4. The spread of the middle half of the data, Q3 – Q1, is the _____ _____.

5. How spread out the numbers in a data set are is its _____.

6. A single number describing how spread out a data set is is a _____ _____ _____.

7. The least or greatest value, far from the others, is an _____.

Write About the Math

Which class scored better overall, and how do the range and the IQR show it?
¿A qué clase le fue mejor en general, y cómo lo muestran el rango y el rango intercuartílico?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Range and Interquartile Range** — *Rango y rango intercuartílico*
- ☐ **Range** — *Rango*
- ☐ **Quartile** — *Cuartil*
- ☐ **Interquartile range** — *Rango intercuartílico*
- ☐ **Variability** — *Variabilidad*

Model: *Equal medians do not make two data sets the same — you still need to know how spread out the scores are.* — Students name a measure of spread rather than another measure of center.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Class A's range is ___ and Class B's range is ___, so Class A's scores are more ___. Class A's IQR is ___ and Class B's IQR is ___, so the middle half of Class B is packed into a ___ stretch of points. Because 75% of Class B scored ___ or more, Class B did better ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.